Staffify Enterprise End-to-End System Guide & Operational Reference Manual

Document Control & Executive Summary

- **Platform Name**: Staffify & StaffifyGo Enterprise HRMS Suite
- **Release Version**: 2.4.0-Enterprise (Production Edition)
- **Architecture**: Distributed Single Page Application (React 19 + Tailwind CSS) with Node.js/Express.js RESTful API, MongoDB Object Modeling (39 Mongoose Schemas), and Google Gemini Multimodal Grounded Intelligence.
- **Audience**: System Administrators, HR Directors, Department Managers, Operations Supervisors, and General Employees.
- **Design System**: Atlassian Enterprise Design System (ADG) compliant with primary `#0052CC`, neutral lozenges, and zero Unicode emojis.

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1. Platform Architecture & Security Foundations

Staffify is architected for enterprise-scale human capital management, ensuring high availability, sub-second query performance, and strict regulatory compliance.

Technical Stack

- **Frontend Layer**: React 19 SPA bundled via Vite, styled using Tailwind CSS and the Atlassian Design System palette (`#0052CC`, `#0747A6`, `#090D14`, `#172B4D`). Icons are rendered via Lucide React.
- **Backend API Layer**: Node.js and Express.js REST application operating under strict JSON contracts, rate limiting, Helmet HTTP security headers, and correlation trace IDs (`X-Request-Id`).
- **Data Persistence Layer**: MongoDB Atlas with 39 specialized Mongoose schemas featuring secondary indexes on `employeeId`, `date`, `jobId`, `candidateId`, and `status`.
- **Artificial Intelligence Engine**: Hybrid dual-tier intelligence consisting of:

1. *Deterministic Semantic Query Engine (DSQE)*: Zero-hallucination parameterized MongoDB aggregator running locally with 100% data privacy. 2. *Google Gemini Multimodal Copilot*: Document vision, resume ATS parsing, and receipt OCR running on `gemini-3.1-flash-lite`.

Security Hardening Standards

- **Zero Server Fingerprinting**: `X-Powered-By` header suppressed across all endpoints.
- **Content Security & Framing**: `X-Content-Type-Options: nosniff` and strict frame ancestry policies blocking clickjacking attacks.
- **NoSQL Injection Defense**: Sanitization middleware automatically strips recursive `$` and `.` operators from URL parameters and request payloads.
- **IDOR Protection**: All record mutations verify authorization context to prevent cross-tenant and horizontal privilege escalations.

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2. Role-Based Access Control (RBAC) Matrix

Staffify categorizes system capabilities across three standardized operational tiers:

Module / Capability	Employee (ESS)	Manager (MSS)	Administrator (HR)
Self-Service Dashboard & Clock-In	Allowed	Allowed	Allowed
Personal Attendance & Regularization	Allowed	Allowed	Allowed
Team Attendance Oversight & Approval	Restricted	Allowed	Allowed
Leave Balance & Leave Application	Allowed	Allowed	Allowed
Subordinate Leave Approval Tray	Restricted	Allowed	Allowed
Employee Master Directory (360 Dossiers)	Read-Only	Read-Only (Team)	Full Control
Payroll Processing & Statutory Lock	Restricted	Restricted	Full Control
Self-Service Payslip View & Print	Allowed	Allowed	Allowed
Goal Setting, OKRs & SMART Refiner	Allowed	Allowed	Allowed
Performance Appraisals & 1-on-1s	Self Review	Calibrate & Grade	System Config
Recruitment Pipeline & Candidate ATS	Restricted	Interviewer Mode	Full Pipeline Control
Expense Claims Submission & OCR	Allowed	Allowed	Allowed
Expense Reimbursement Approval Tray	Restricted	Allowed	Allowed
Shift Schedule & Shift Swapping	Allowed	Team Roster Matrix	Global Config
IT Service Desk Tickets & Assets	Allowed	Triage Support	Inventory Control
Project Timesheets Logging	Allowed	Review & Approve	Financial Audit
Executive Headcount Analytics	Restricted	Department Scope	Organization Scope
Resignation & Offboarding Clearances	Self Resign	Department Clearance	Full & Final Payout
AI Copilot & Multimodal Studio	Role-Scoped	Role-Scoped	Full System Access
Tamper-Evident Audit Trail Logs	Restricted	Read-Only	Audit Export
Outbound Webhook Bridge Gateway	Restricted	Restricted	Configure & Dispatch

3. Authentication & Session Lifecycle

The authentication gateway provides single-credential login with instant role preview presets for rapid enterprise testing and deployment.

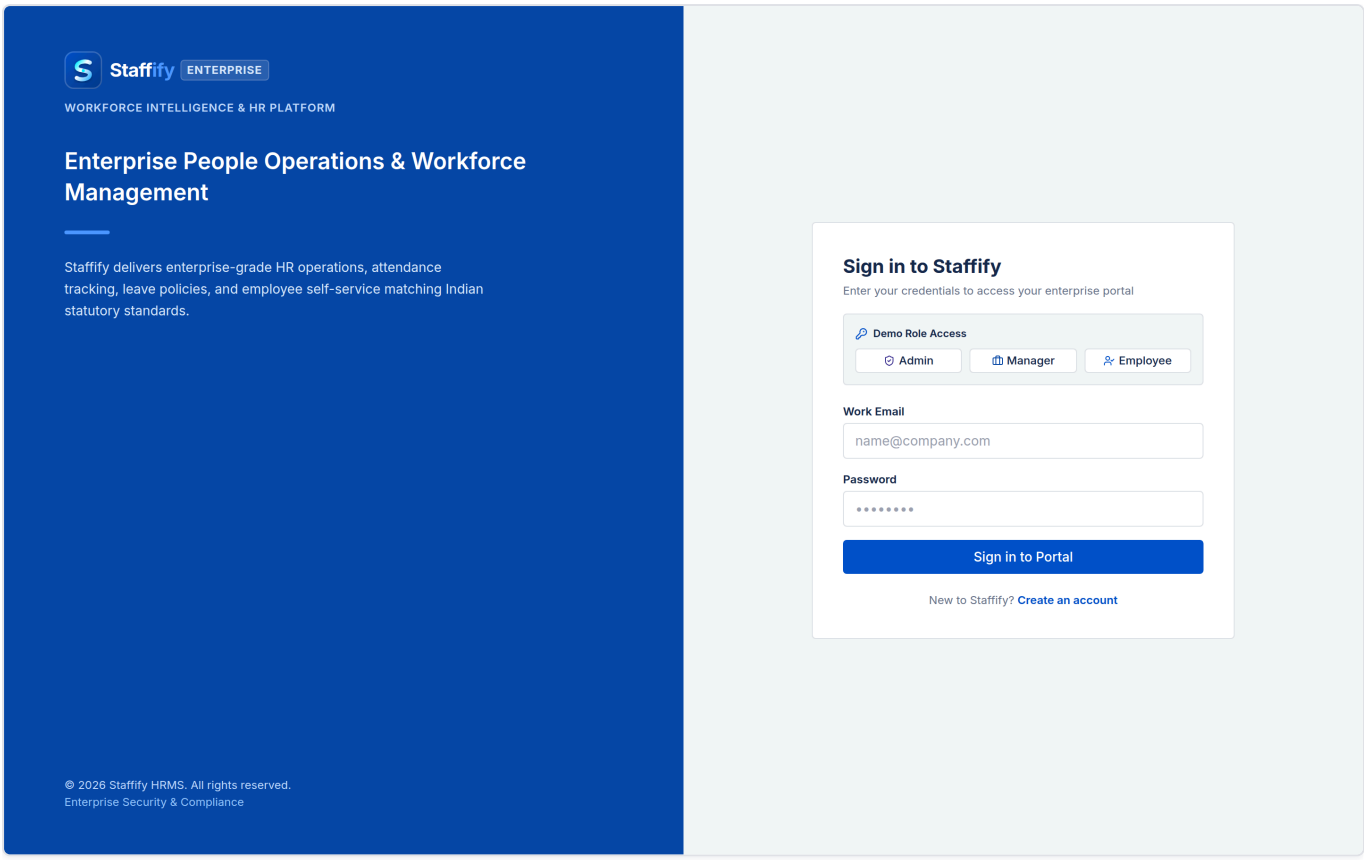


Figure: Staffify Unified Enterprise Login Gateway

Key Features

- **JSON Web Token (JWT) Security**: Issues cryptographically signed 24-hour access tokens containing user identifiers, employee IDs, and verified system roles.
- **BCrypt Password Hashing**: Passwords stored using 10 salt rounds with adaptive computational work factor.
- **Demo Role One-Click Switchers**: Immediate access to verified demo accounts:

- Administrator: `admin@staffify.com` / `Admin@123` - Manager: `manager@staffify.com` / `Manager@123` - Employee: `employee@staffify.com` / `Employee@123`

Module 01: Workforce Command Center & Executive Dashboard

The central landing command center synthesizes workforce KPIs, daily attendance widgets, real-time announcements, and pending action items.

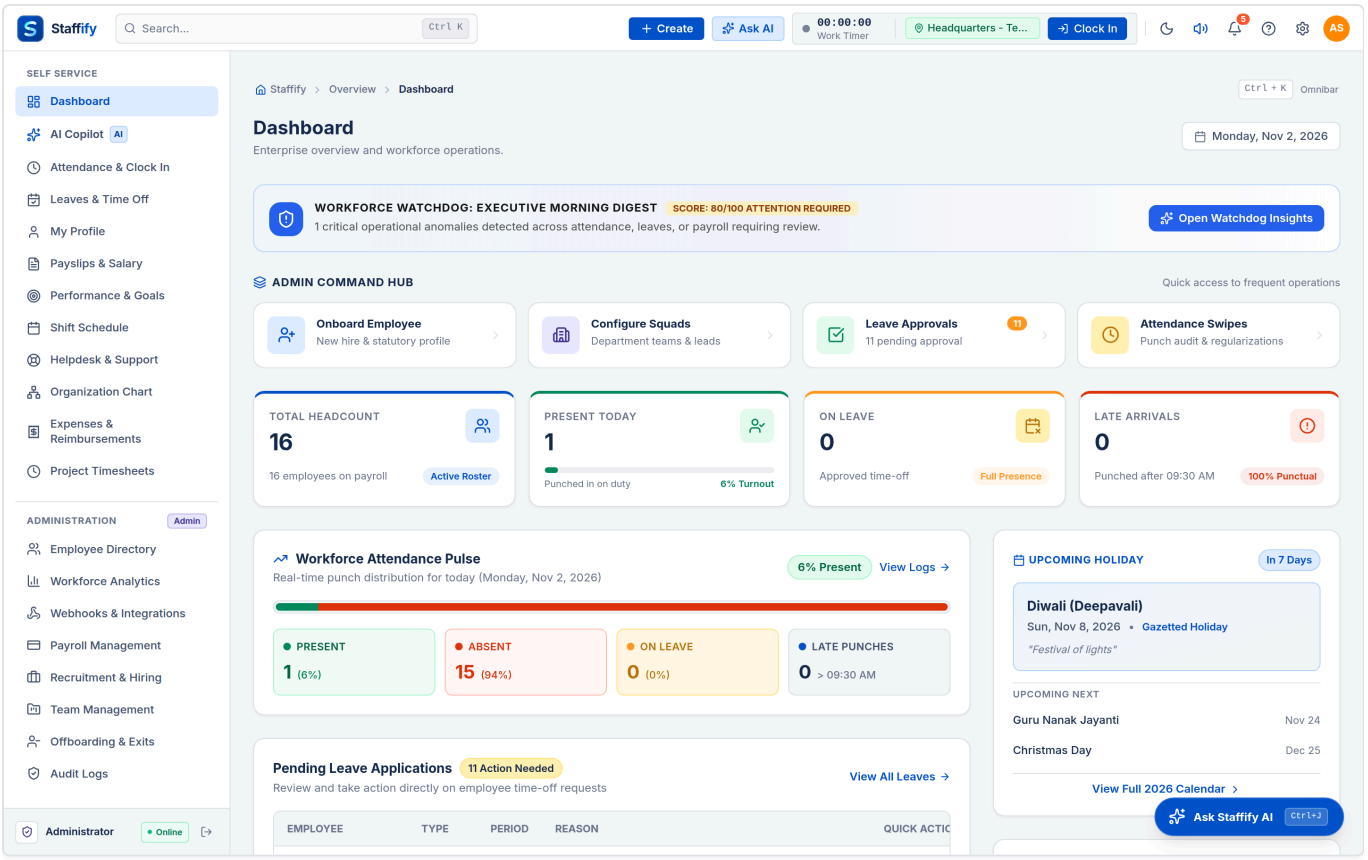


Figure: Workforce Command Center & Dashboard

Key Functional Capabilities

1. **Real-Time Attendance Clock-In Widget**: - Single-click **Clock In** and **Clock Out** action dispatcher with live time tracking. - Automatically registers current IP address, GPS coordinate perimeters, and biometric device bindings. 2. **Executive Metric Tiles**: - Displays Total Headcount, Today's Present Workforce, Pending Leave Applications, and Active Support Tickets. 3. **Interactive Quick Actions Toolbar**: - Deep-linked shortcuts to Apply for Leave, Submit Reimbursement Claims, Log IT Tickets, or Launch AI Copilot. 4. **Broadcast & Announcement Feed**: - Organization-wide notices published by HR leadership with priority banners and delivery timestamps.

Module 02: Attendance Management, Geofencing & Biometric Logs

Comprehensive daily timekeeping and physical workplace tracking combining mobile GPS geofencing, IP whitelisting, and eSSL/ZKTeco push biometric terminals.

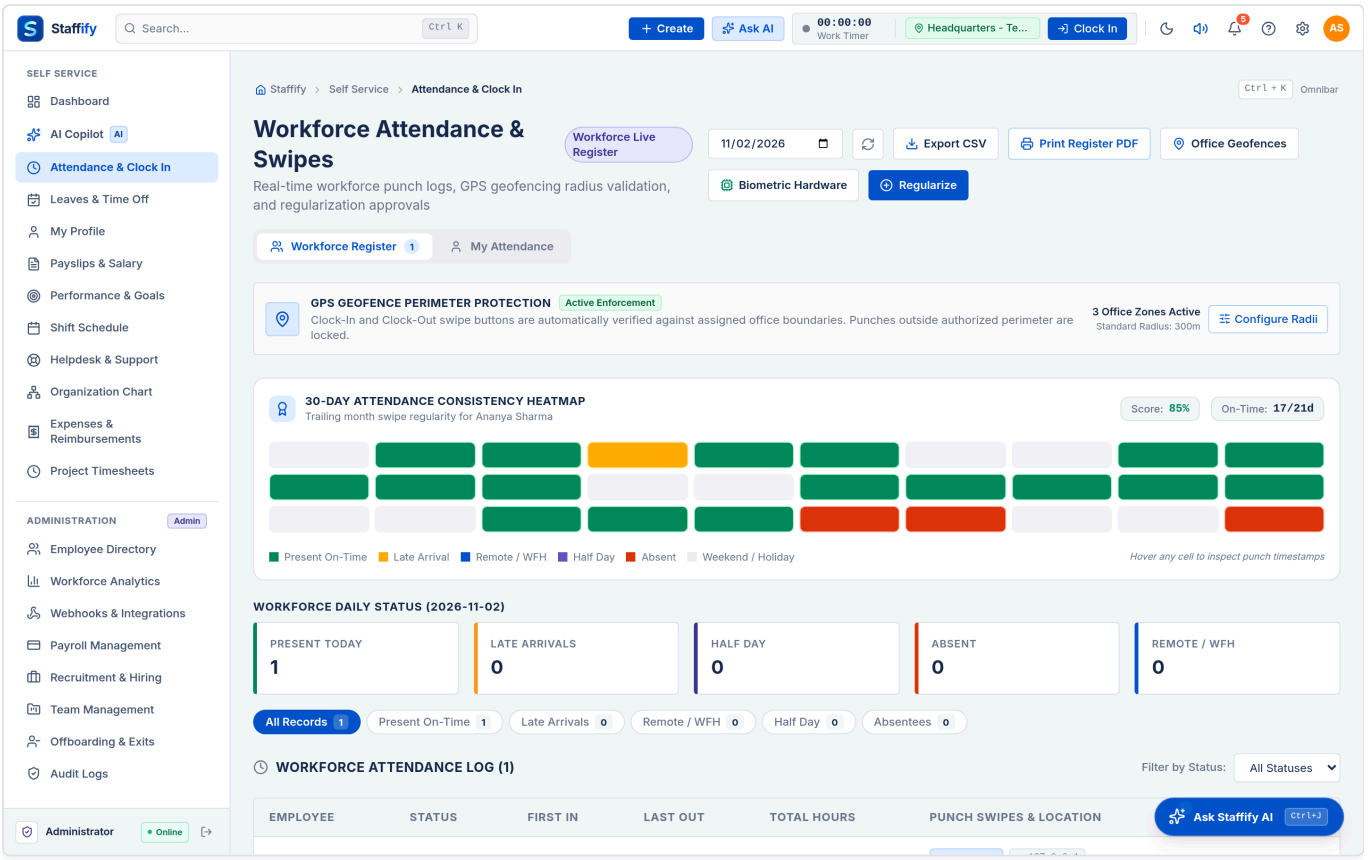


Figure: Attendance Register & Geofencing Monitor

Daily Work Hours & Status Rules

- **Present**: First clock-in on or before 09:30 AM with at least 8.0 logged working hours.
- **Late Check-In**: First clock-in after 09:30 AM (marked with warning status lozenge).
- **Half Day**: Total daily logged duration between 4.0 and 8.0 hours.
- **Absent**: Under 4.0 hours recorded with no approved leave or regularization.

Geofence Enforcement & Mobile Validation

- Automatically measures Euclidean and Haversine distance between mobile clock-in GPS coordinates and corporate office coordinates (e.g. Bangalore HQ: `12.9716, 77.5946` within a 150-meter radius).
- Outside-perimeter punches flag an `Unverified Location` anomaly for supervisor review.

Embedded AI Regularization Reason Drafter

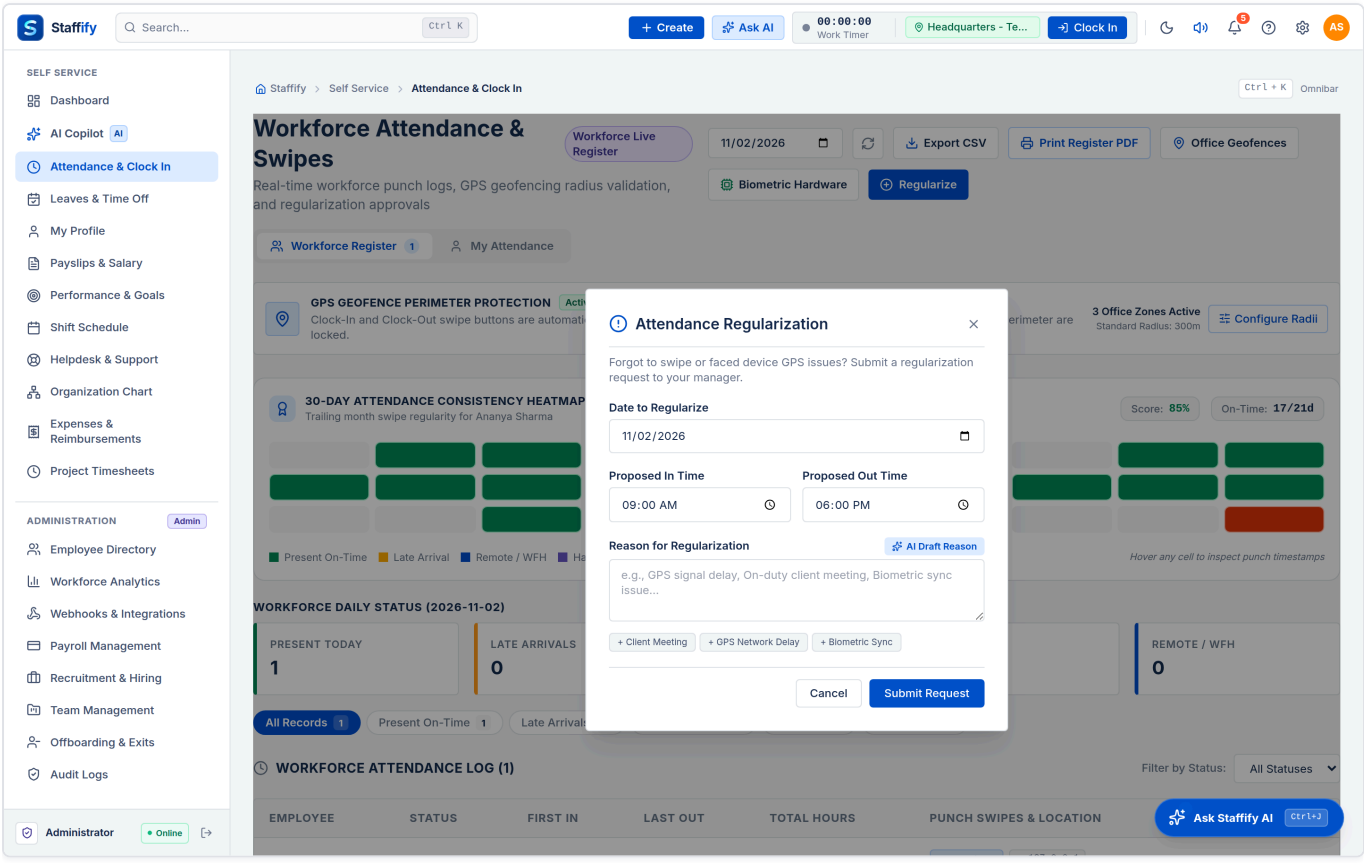


Figure: AI Attendance Regularization Modal

When an employee experiences turnstile delays or remote client travel:
1. Click **Regularize** on the disputed attendance row.
2. Click **AI Draft Reason** to automatically synthesize a legally sound explanation.
3. Or select quick corporate preset pills:
- *GPS geofence radius deviation due to mobile network carrier latency.*
- *Biometric terminal hardware synchronization delay at entrance turnstile.*
- *Official duty client travel with reporting manager pre-authorization.*
4. Submit the request directly to the manager's approval tray.

Module 03: Leave Management & Statutory Sandwich Rule Enforcement

Unified annual, casual, sick, and compensatory off management adhering strictly to statutory labour frameworks and corporate leave policies.

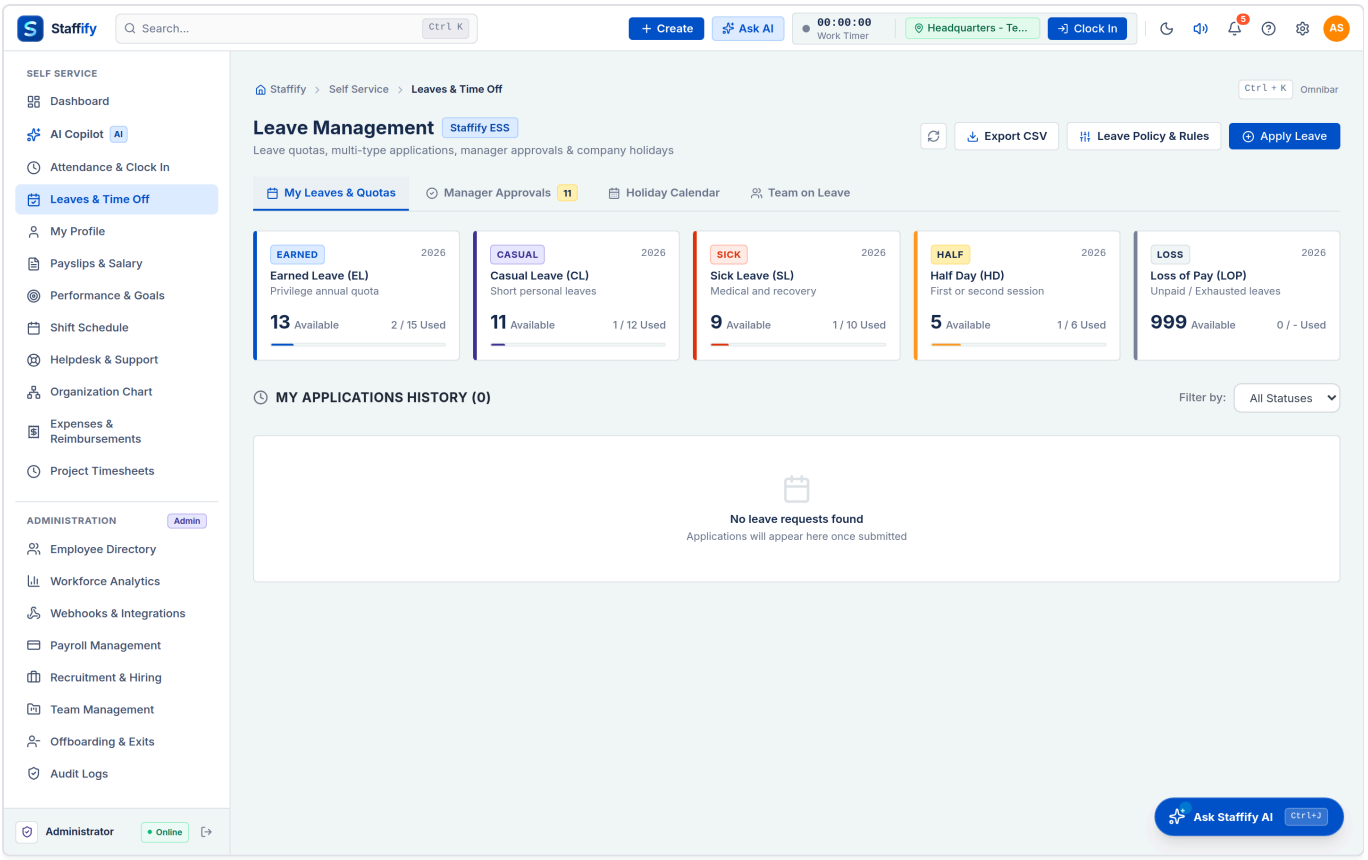


Figure: Leave Quotas & Approval Management

The Statutory Sandwich Rule

Under Staffify's automated sandwich rule engine:

- If an employee takes leave on Friday and Monday (bridging a weekend), Saturday and Sunday are automatically counted as leave days if the company's sandwich policy is enabled.
- Prevents uncoordinated long-weekend absenteeism while maintaining mathematical balance tracking.

Embedded AI Leave Reason Drafter

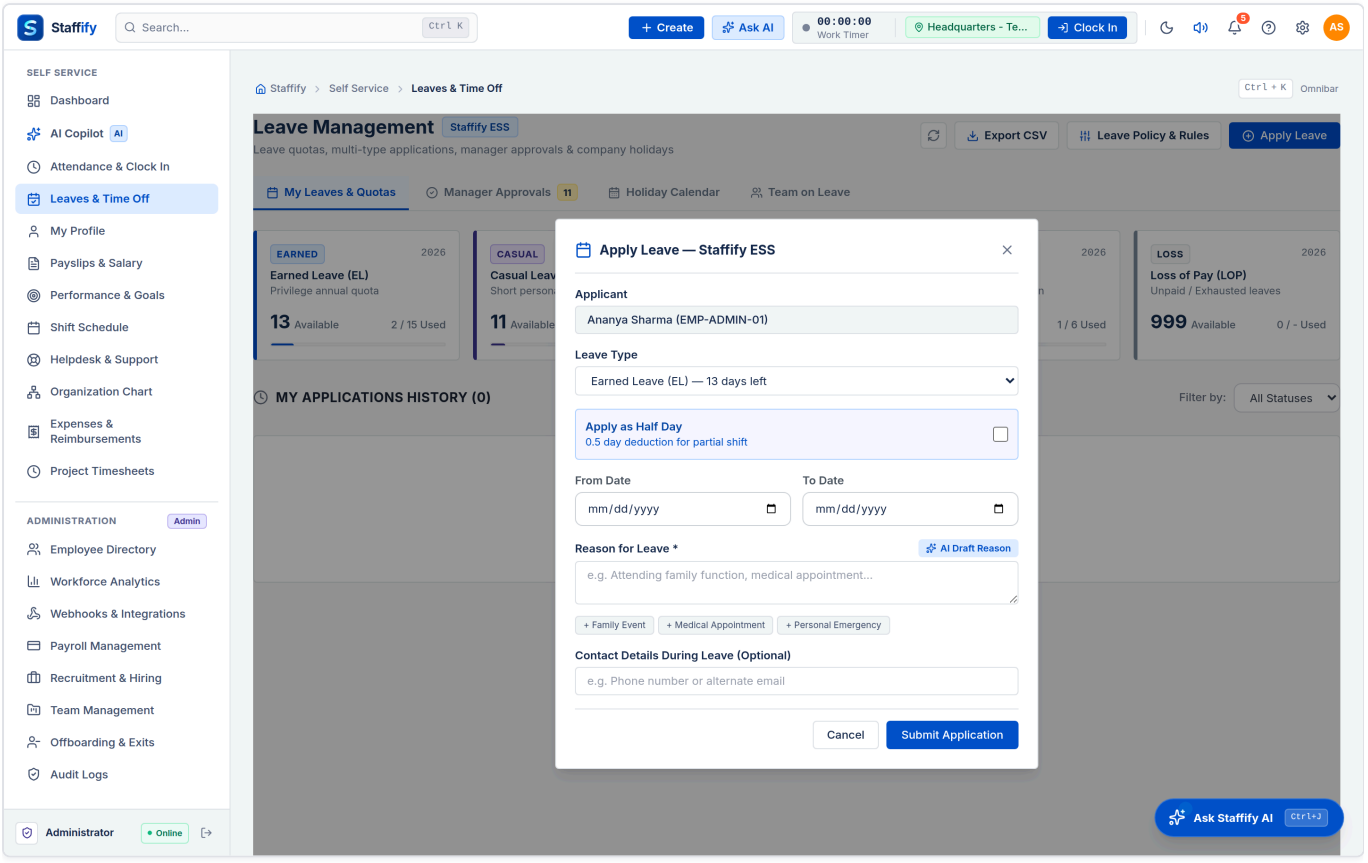


Figure: Apply Leave Modal with AI Reason Drafter

- **One-Click Professional Drafter**: Formulates courteous, unambiguous reasons for planned absences.
- **Contextual Scenario Pills**:

- Personal family commitment requiring planned absence; reachable via phone if urgent. - Scheduled medical check-up and restorative wellness rest. - Out-of-town domestic and municipal documentation work.

- **Live Balance Projection**: Dynamically computes remaining balance for Casual (CL), Sick (SL), and Earned (EL) leave quotas prior to submission.

Module 04: Employee Directory & 360-Degree Personnel Dossiers

The authoritative employee record system storing personal, emergency, statutory, bank, and organizational hierarchy attributes.

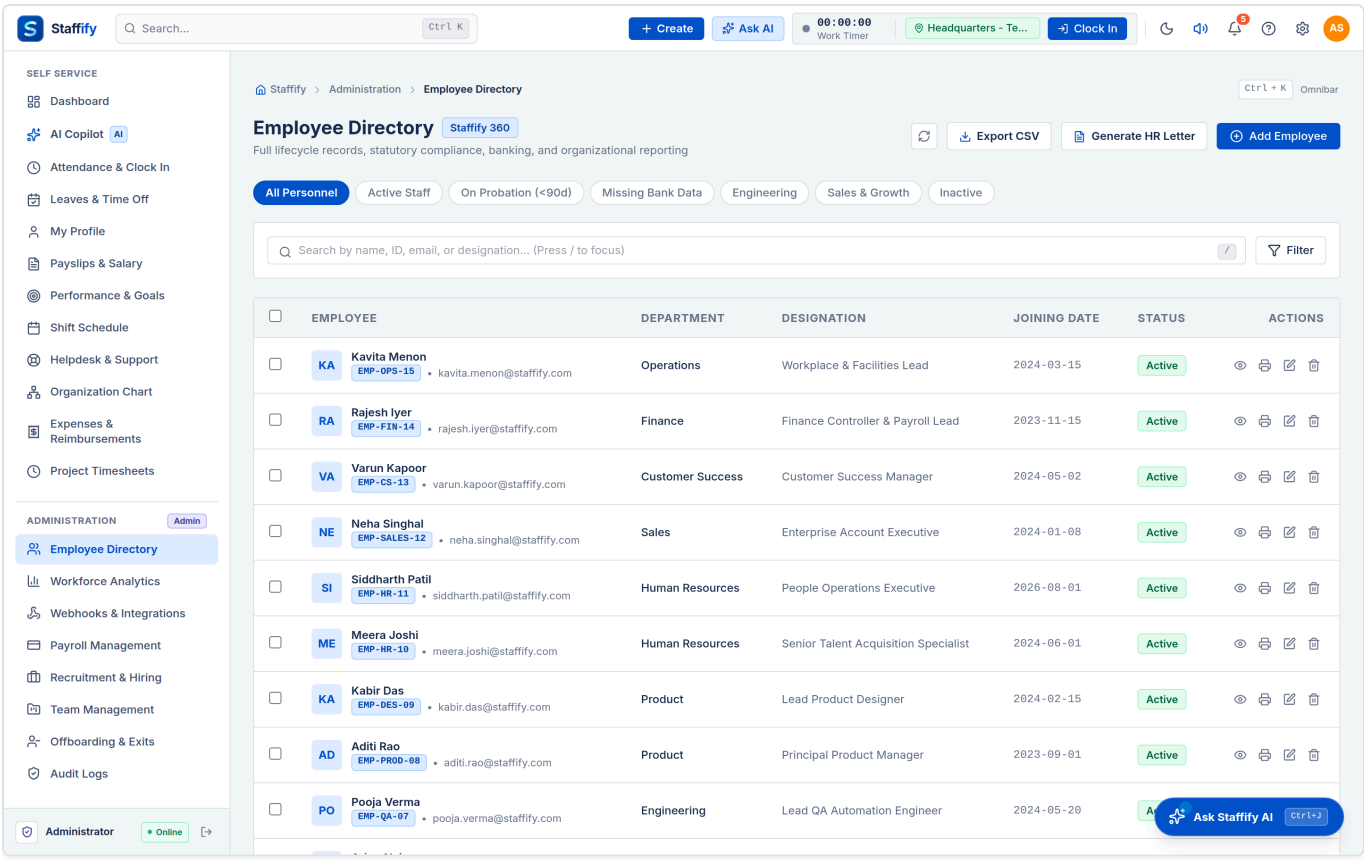


Figure: Employee Master Directory

Key Capabilities

- **Multi-Filter Search**: Search across employee names, designations, departments, and active statuses with instantaneous debounced filtering.
- **Direct Dossier Print Action**: Print a standardized vector personnel dossier PDF directly from the table row via the printer icon.
- **Bulk Action Toolbar**: Select multiple employees for department reassignment, status updates, or bulk data export to CSV.

The 360-Degree Employee Dossier Modal

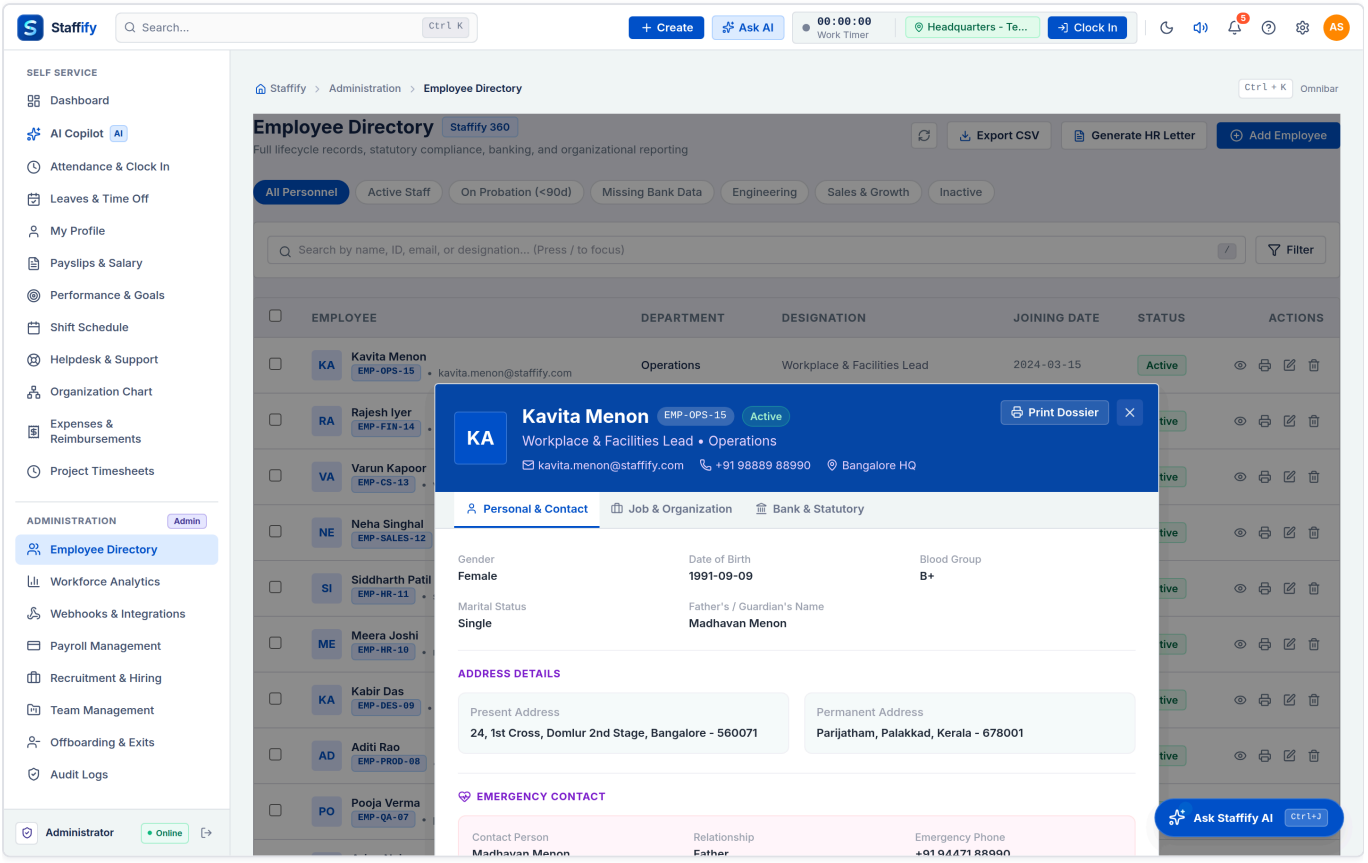


Figure: Employee 360-Degree Personnel Dossier Modal

- Tabbed executive inspection of **Personal**, **Emergency Contacts**, **Bank Account**, **Statutory (PAN/Aadhaar/UAN/PF)**, and **Reporting Structure**.
- Features an executive **Print Dossier** action that triggers the zero-popup vector print engine to generate an official 3-page personnel record file.

Module 05: Payroll Execution & Statutory Compliance (EPF, ESIC, PT, TDS)

A mathematically validated Indian payroll processing engine incorporating statutory rules and one-click salary disbursement lock.

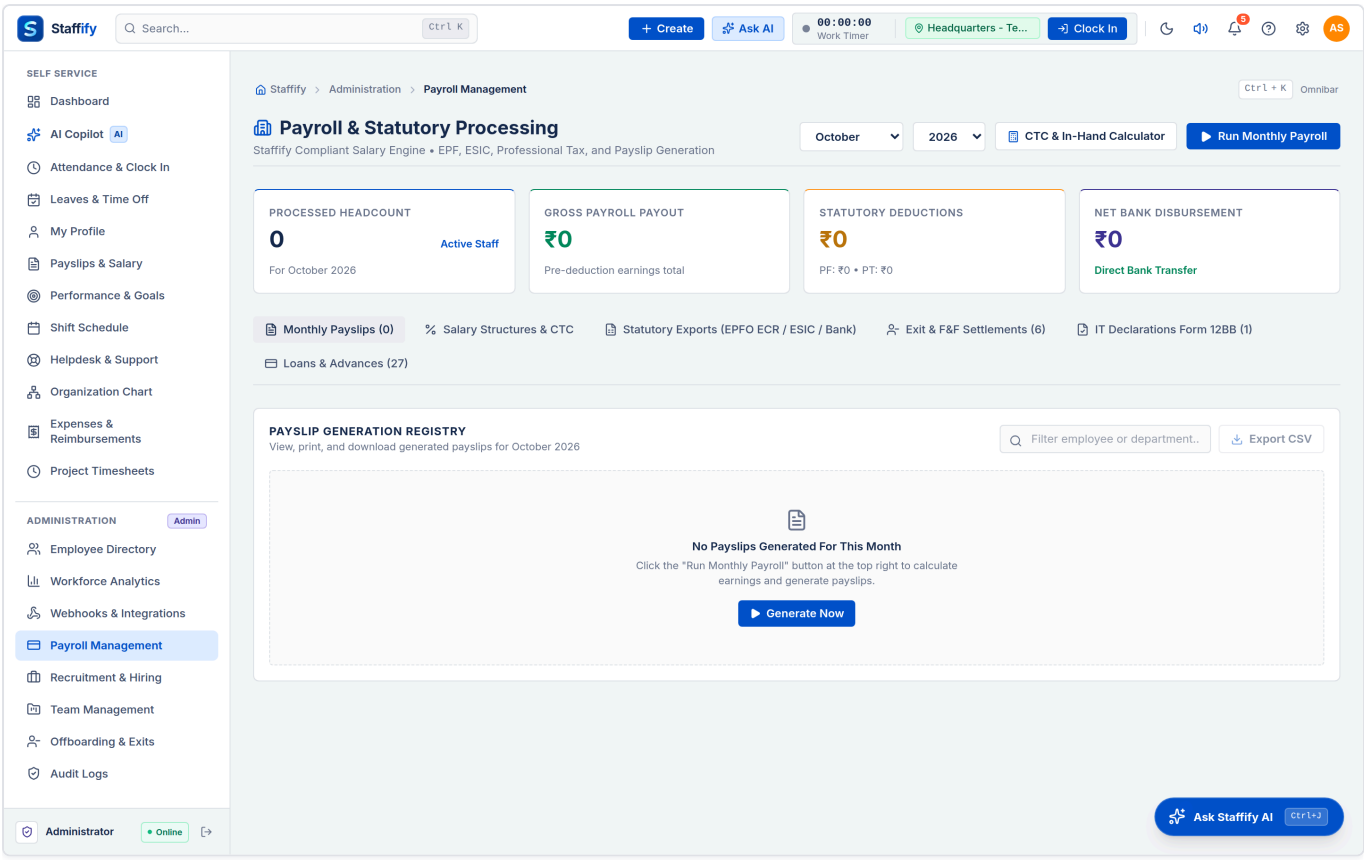


Figure: Payroll Processing & Statutory Compliance Monitor

Formulaic Salary Breakdown

- **Basic Salary**: 50% of Total Fixed CTC.
- **House Rent Allowance (HRA)**: 50% of Basic Salary (for metro locations).
- **Special Allowance**: Balancing component (`CTC - Basic - HRA`).
- **Employee Provident Fund (EPF)**: 12% of Basic Salary (statutory cap applicable).
- **Employee State Insurance (ESIC)**: 0.75% of Gross Pay (for employees earning up to INR 21,000/month).
- **Professional Tax (PT)**: INR 200/month (statutory state slab).
- **TDS / Income Tax**: Dynamically calculated based on Old vs New regime tax declarations.

Administrative Controls

- **Run Monthly Payroll**: Automatically aggregates working days, unpaid leaves (LWP), and approved loan EMI deductions to produce finalized payroll drafts.
- **Statutory Returns Export**: 1-click download of EPFO Electronic Challan cum Return (ECR) text files, ESIC contribution statements, and NEFT/RTGS bank disbursement sheets.

Module 06: Employee Self-Service Payslips & Salary Statements

Secure portal for employees to review historical monthly compensation, statutory deductions, tax withholdings, and download authentic salary statements.

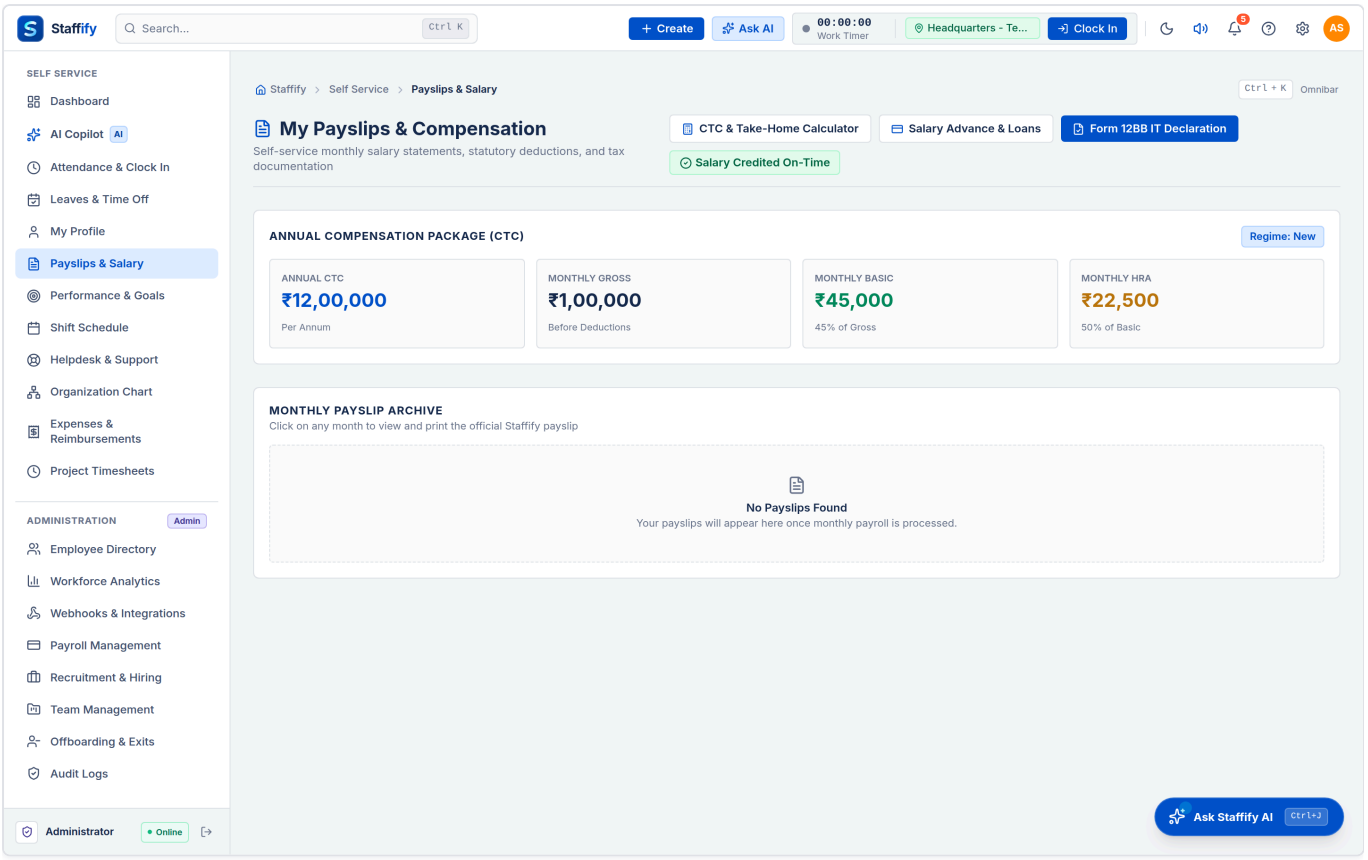


Figure: Employee Payslip Self-Service Portal

Features

- **Visual Breakdown**: Interactive cards showing Net Pay, Gross Earnings, and Total Deductions for every historical cycle.
- **Zero-Popup Vector Print**: Launches browser print dialog to print or "Save as PDF" with pristine vector fonts, letterhead styling, and net pay in words (e.g. *INR One Lakh Forty-Three Thousand Only*).

Module 07: Performance Management, OKRs & AI SMART Refiner

Continuous performance management aligning individual key results with organizational Objectives and Key Results (OKRs).

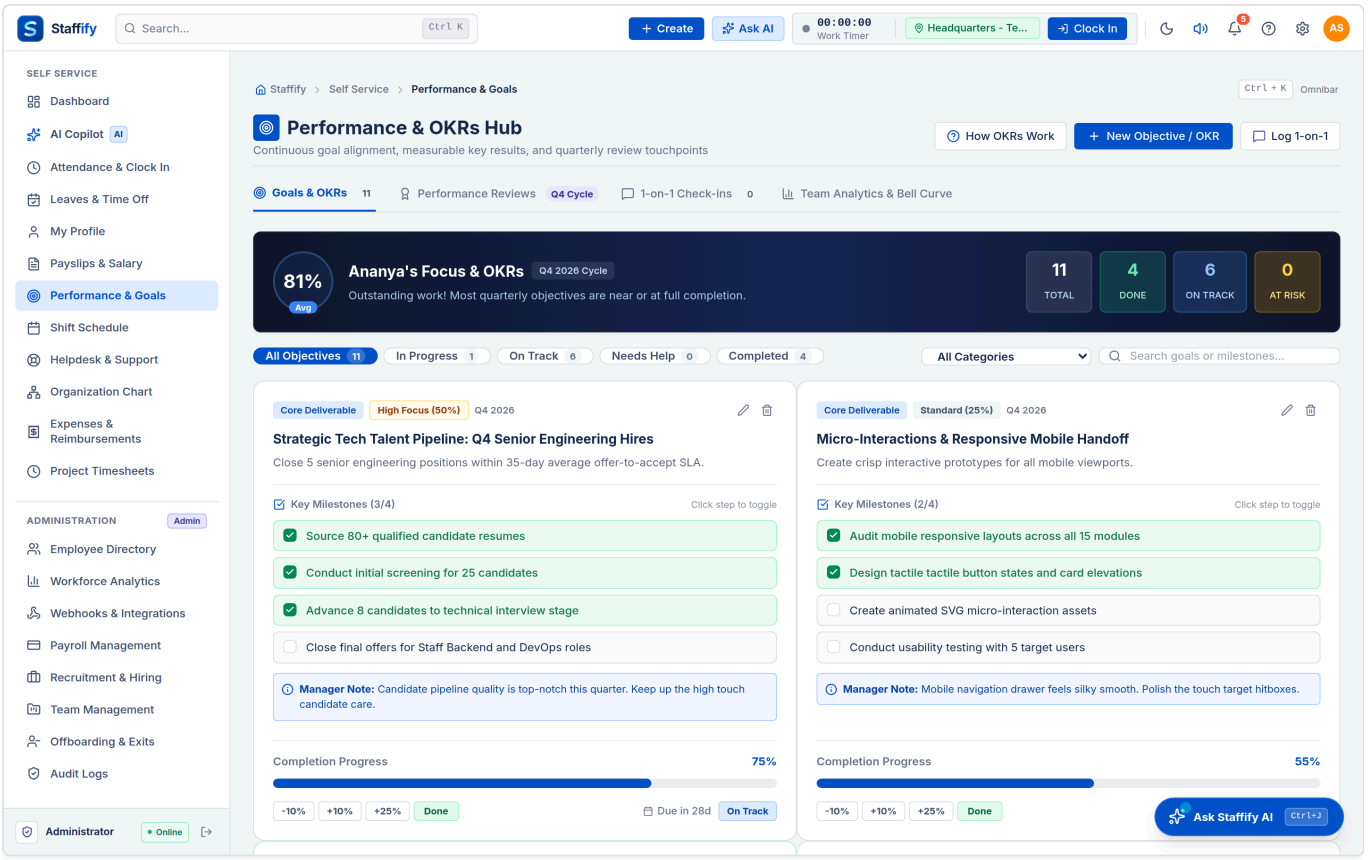


Figure: Performance Objectives & OKRs

Key Capabilities

- **OKR Progress Sliders**: Drag-and-drop progress percentage tracking with visual completion statuses (Not Started, In Progress, On Track, At Risk, Completed).
- **Quarterly Appraisals**: Multi-tier review workflow supporting self-appraisal submissions, manager evaluations, and normalized final ratings.
- **Continuous 1-on-1 Logs**: Action item tracker for manager-employee coaching syncs.

Embedded AI SMART Goal Refiner

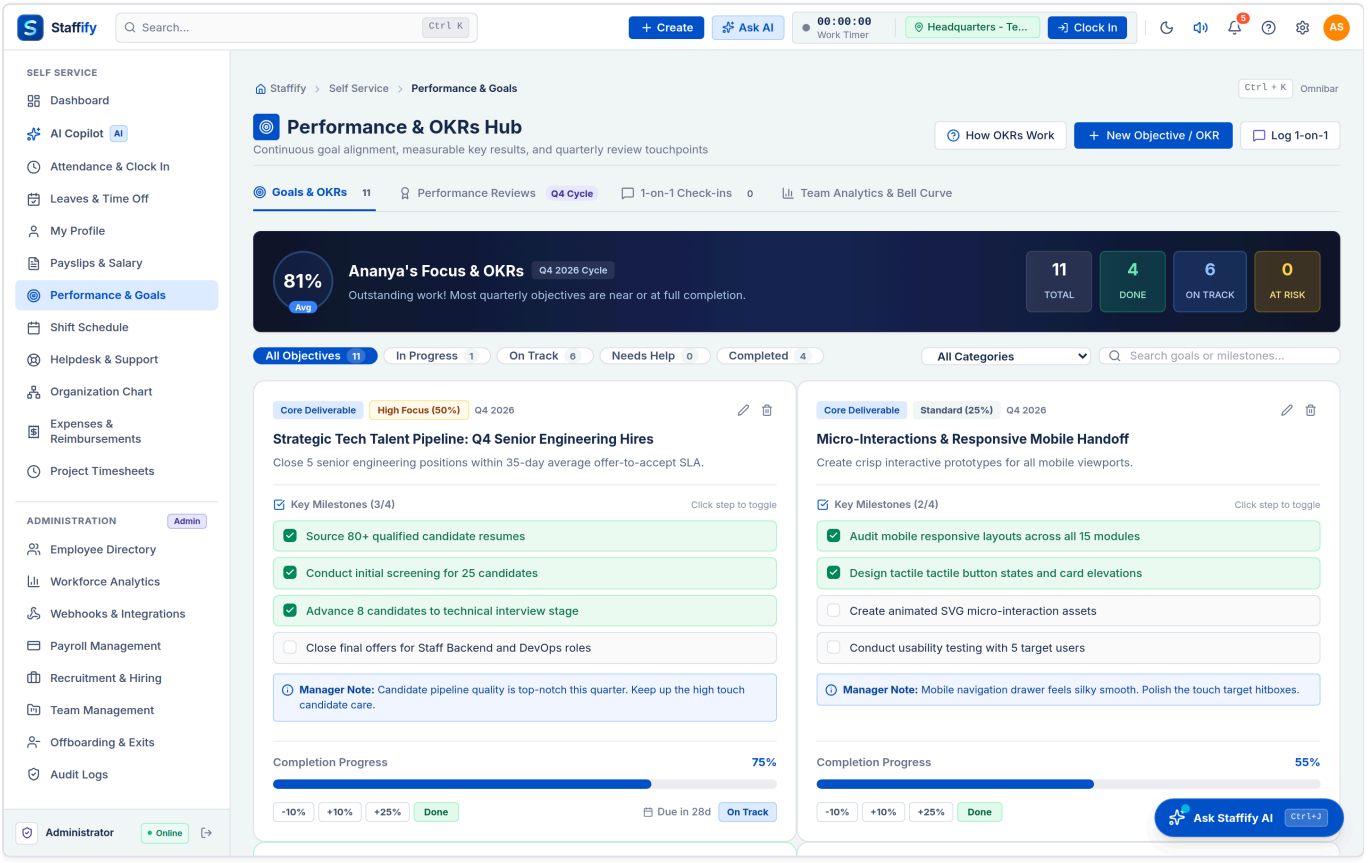


Figure: AI SMART Goal Refiner Modal

Transform vague, non-actionable goals into structured OKRs:
1. Type a rough idea (e.g. *Speed up app* or *Learn cloud*).
2. Click **AI SMART Refiner**.
3. The engine automatically formulates:
- A crisp, inspiring **Objective Title** (e.g. *Accelerate Platform Query Execution & Attain Sub-200ms API Response Benchmark*).
- Three **Measurable Key Milestones** (e.g. *Index high-traffic MongoDB collections*, *Implement in-memory caching tier*, *Reduce bundle transfer payload by 35%*).
- Appropriate **Category** and **Focus Level Weightage**.

Module 08: Recruitment ATS, Job Requisitions & AI Interview Kits

End-to-end Applicant Tracking System (ATS) managing job postings, resume parsing, candidate stages, and interview scorecards.

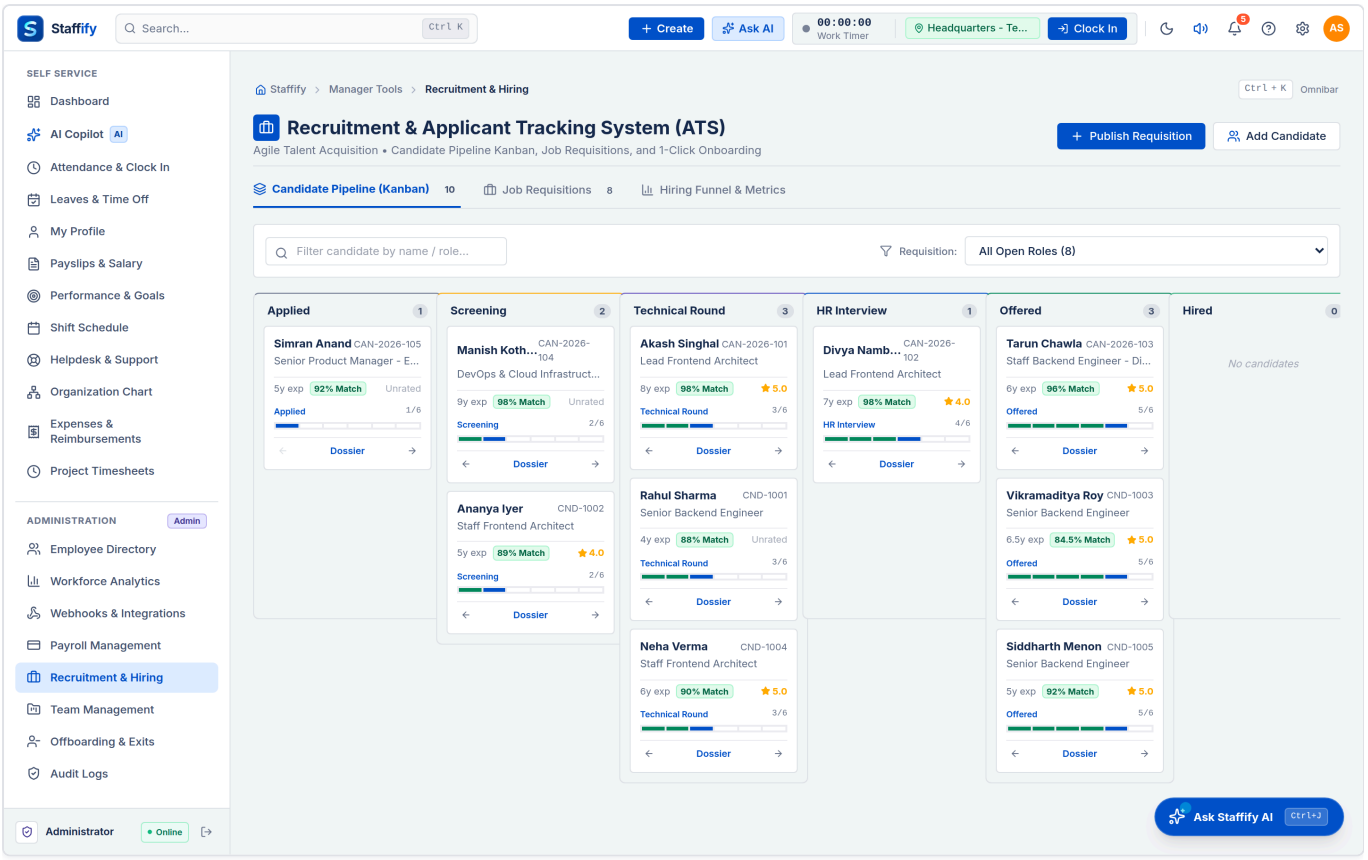


Figure: Recruitment ATS & Candidate Pipeline

Visual Hiring Pipeline Stages

Candidates progress smoothly through seven structured stages: `Applied` $\rightarrow$ `Screening` $\rightarrow$ `Technical Round` $\rightarrow$ `HR Interview` $\rightarrow$ `Offered` $\rightarrow$ `Hired` $\rightarrow$ `Rejected`.

Embedded AI Multimodal Resume Parser & Interview Kit

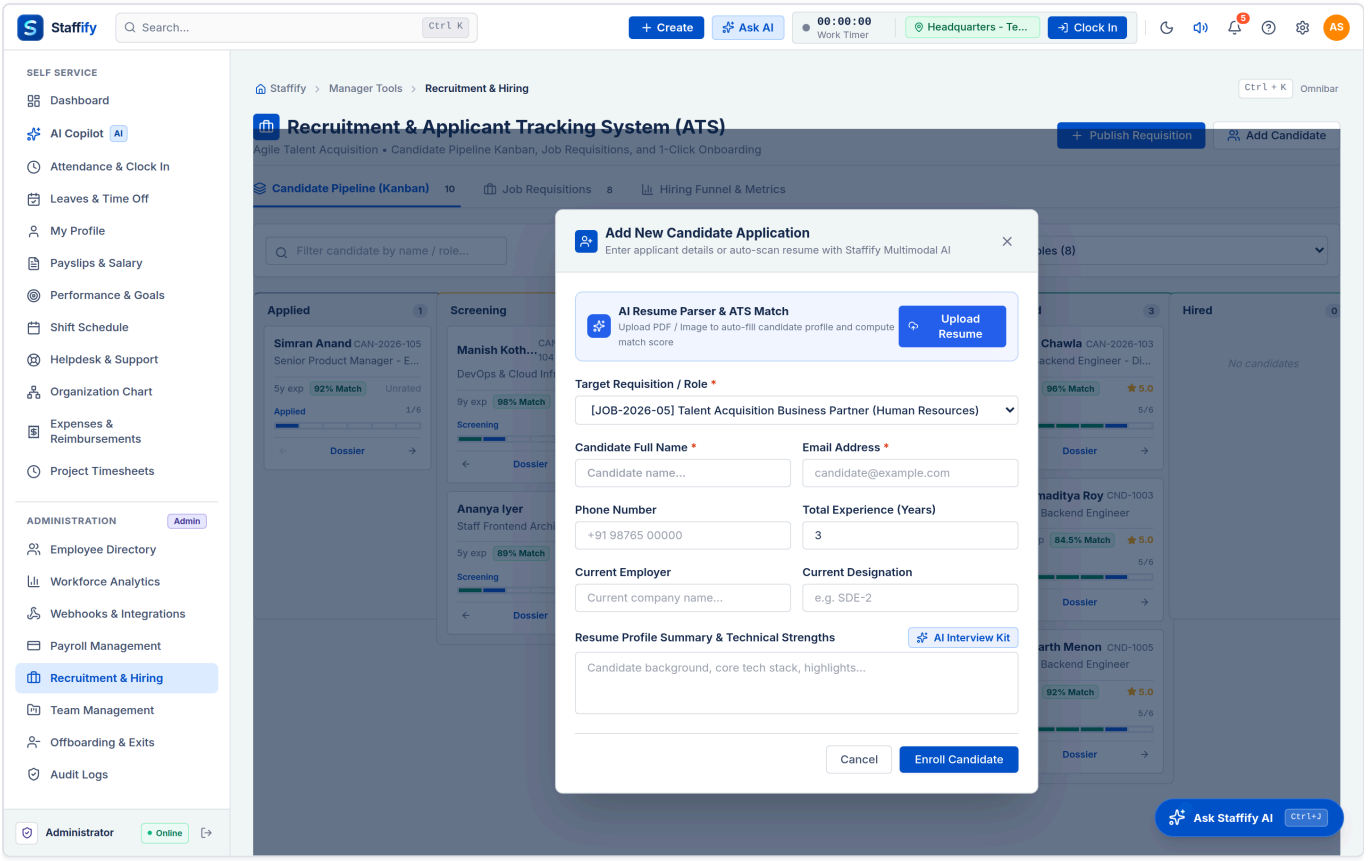


Figure: Candidate Application Modal with AI Interview Kit

- **Multimodal Resume OCR**: Upload a candidate PDF resume to auto-populate contact details, years of experience, current employer, and calculate an ATS match score.
- **AI Interview Kit Generator**: Synthesizes 3 customized technical and behavioral interview questions tailored to the requisition title and experience level.
- **Scorecard Feedback Assistant**: Inside the candidate dossier, interviewers can click **AI Assessment Drafter** to quickly synthesize balanced, structured interview notes.

Module 09: Expense Claims, Policy Compliance Guard & Reimbursements

Automated corporate expense management covering travel, client entertainment, equipment, and office consumables.

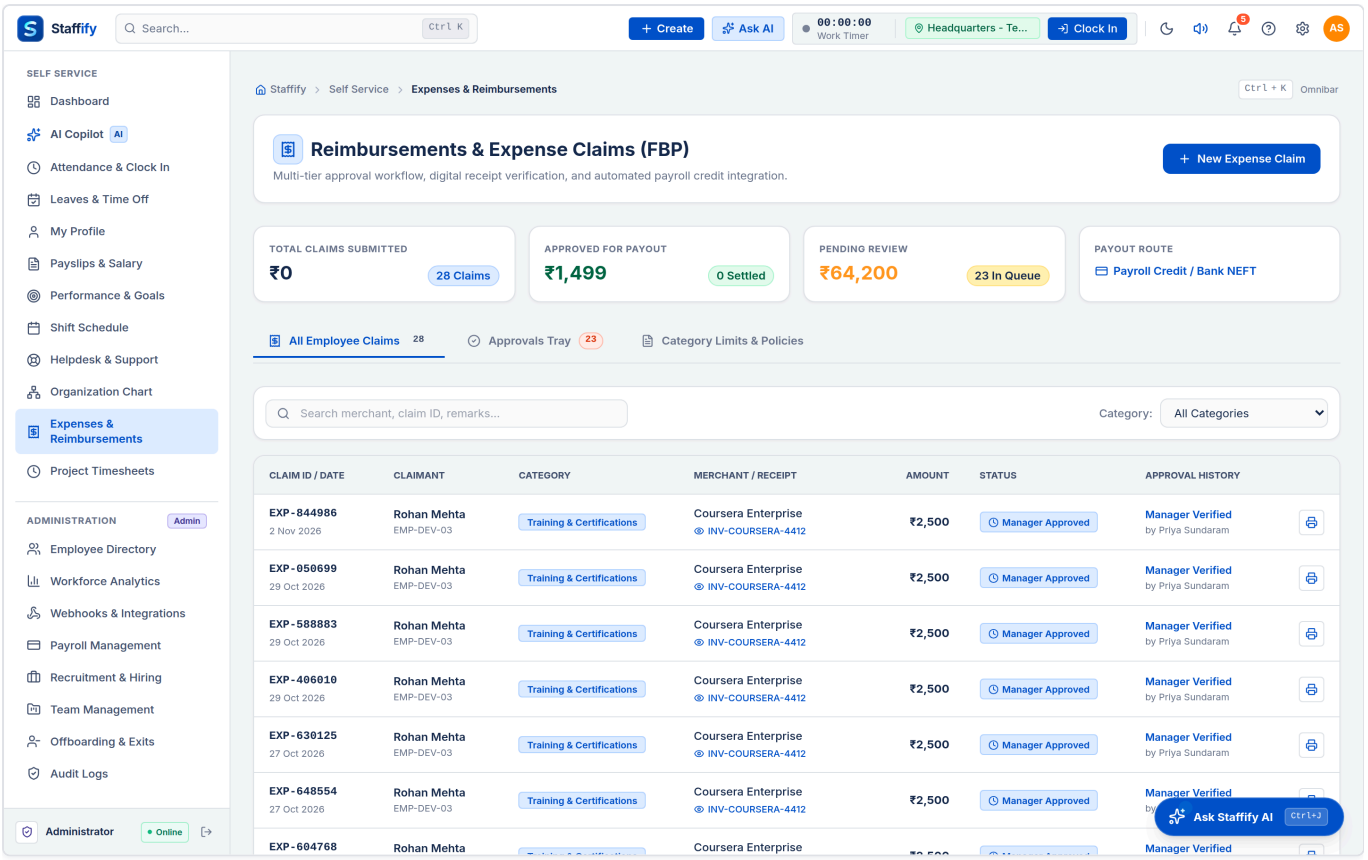


Figure: Expense Reimbursement Claims Tray

Embedded AI Policy Compliance Guard

When an employee enters claim details in the submission modal:

- The system checks line-item amounts against corporate category thresholds (e.g. INR 2,500 for Food, INR 5,000 for Local Conveyance, INR 15,000 for Travel).
- Displays real-time warnings if a claim exceeds policy limits or requires senior management override.

Printable Reimbursement Vouchers

- Click **Print Voucher PDF** on any approved or pending claim.
- Generates a formal A4 reimbursement voucher complete with tax breakdown, category codes, policy compliance seal, and employee/approver signature lines.

Module 10: Squad Shift Rostering & Swap Request Workflows

24/7 squad scheduling system for engineering on-call, technical support, operations, and sales squads.

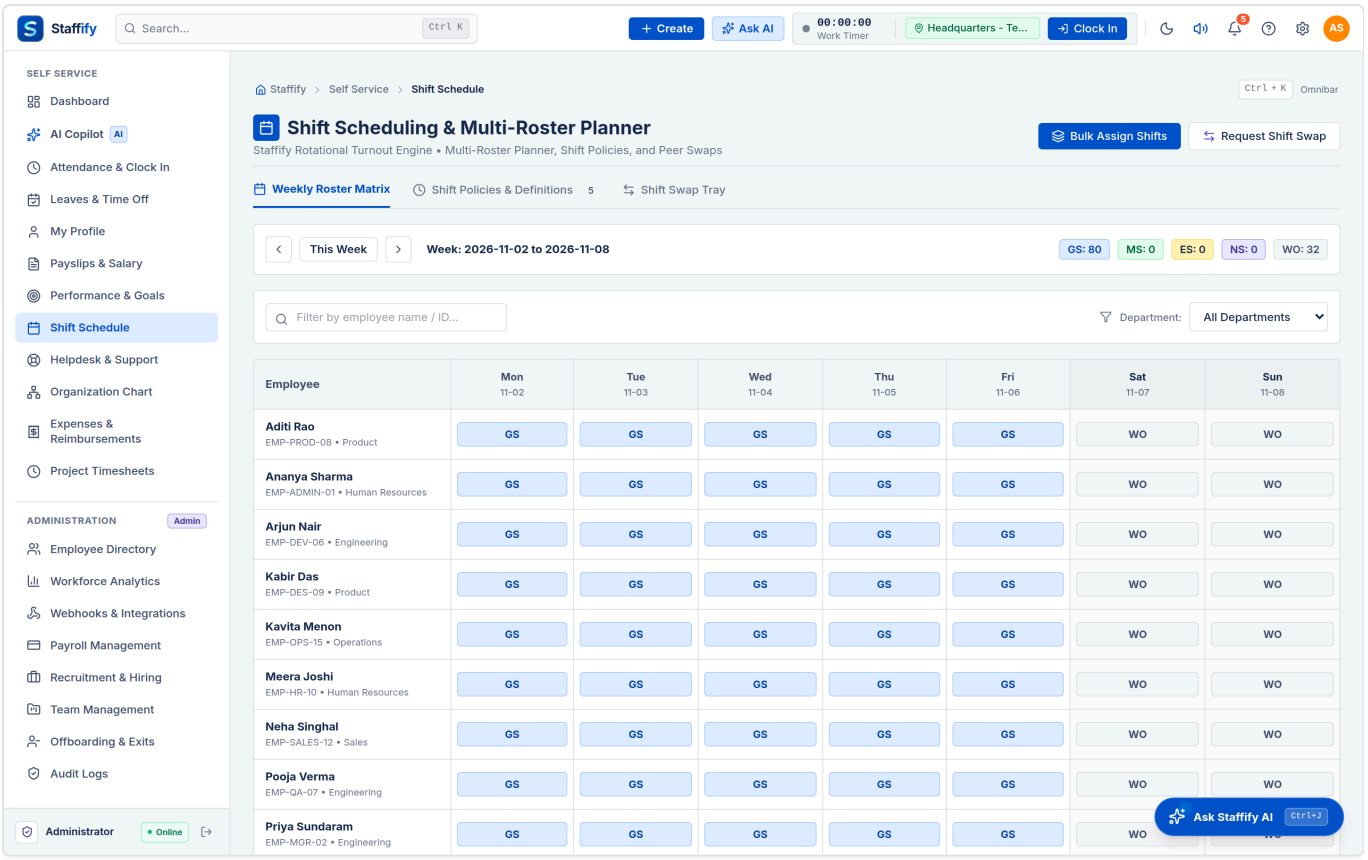


Figure: Squad Shift Rostering Matrix

Features

- **Shift Templates**: Pre-configured shifts including General (09:00 - 18:00), Morning (06:00 - 15:00), Afternoon (14:00 - 23:00), and Night (22:00 - 07:00).
- **Roster Matrix Grid**: Weekly grid view mapping team members to daily shift assignments with conflict detection.
- **Shift Swap Requests**: Employees can request peer shift swaps with automatic manager approval routing.

Module 11: Helpdesk Service Tickets & IT Asset Management

Internal IT and HR helpdesk ticketing system integrated directly with hardware inventory allocation.

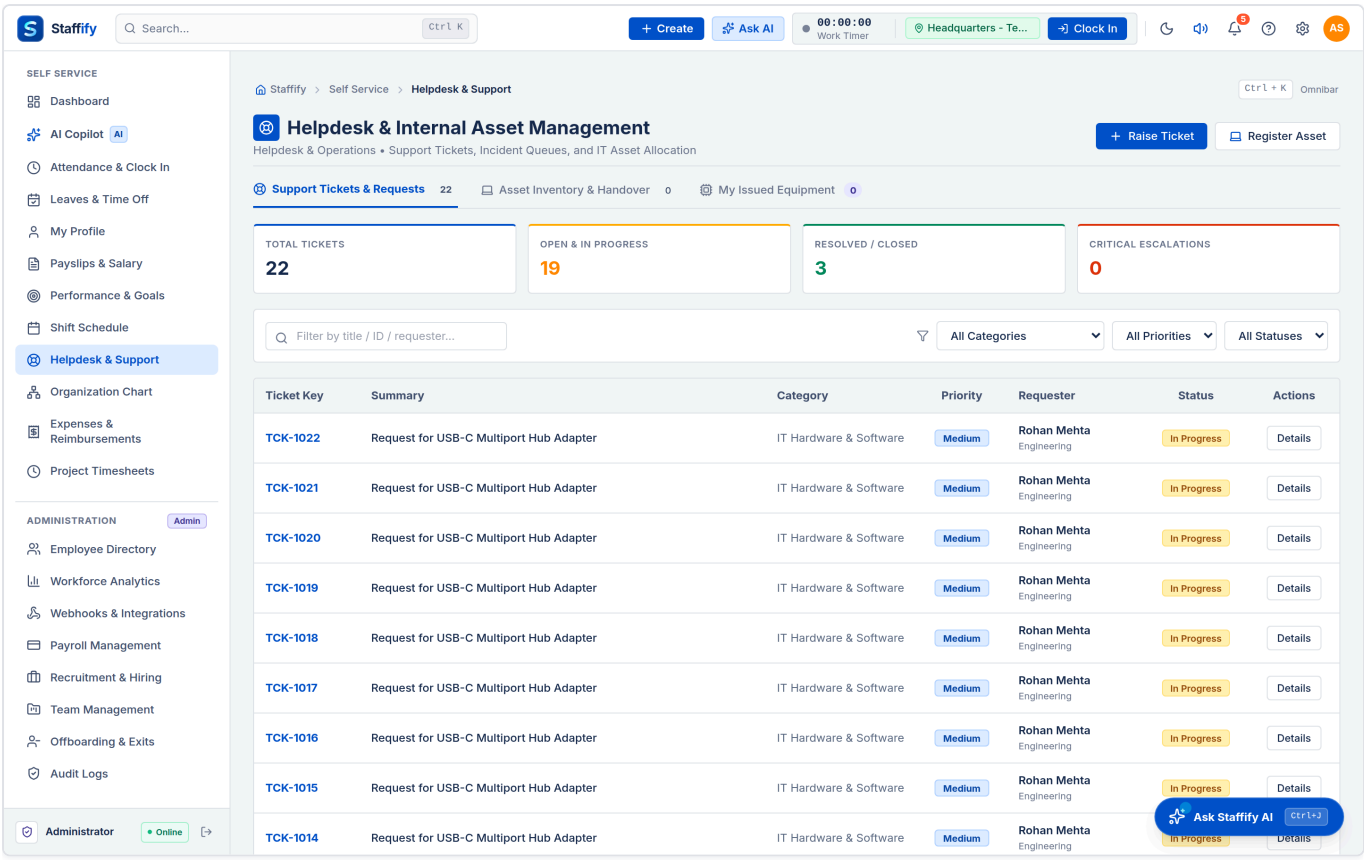


Figure: Helpdesk Tickets & Asset Management

Key Capabilities

- **Priority-Based Ticketing**: Log Low, Medium, High, or Critical tickets with SLA countdown timers.
- **Hardware Asset Allocation**: Tracks assigned laptops, monitors, access badges, and peripherals with serial numbers, warranty dates, and condition logs.
- **Status Triage**: Filter tickets by `Open`, `In Progress`, `Resolved`, and `Closed` with technician notes.

Module 12: Client Timesheets & Project Hours Tracking

Weekly project time allocation tracker for client billing, internal R&D tracking, and productivity monitoring.

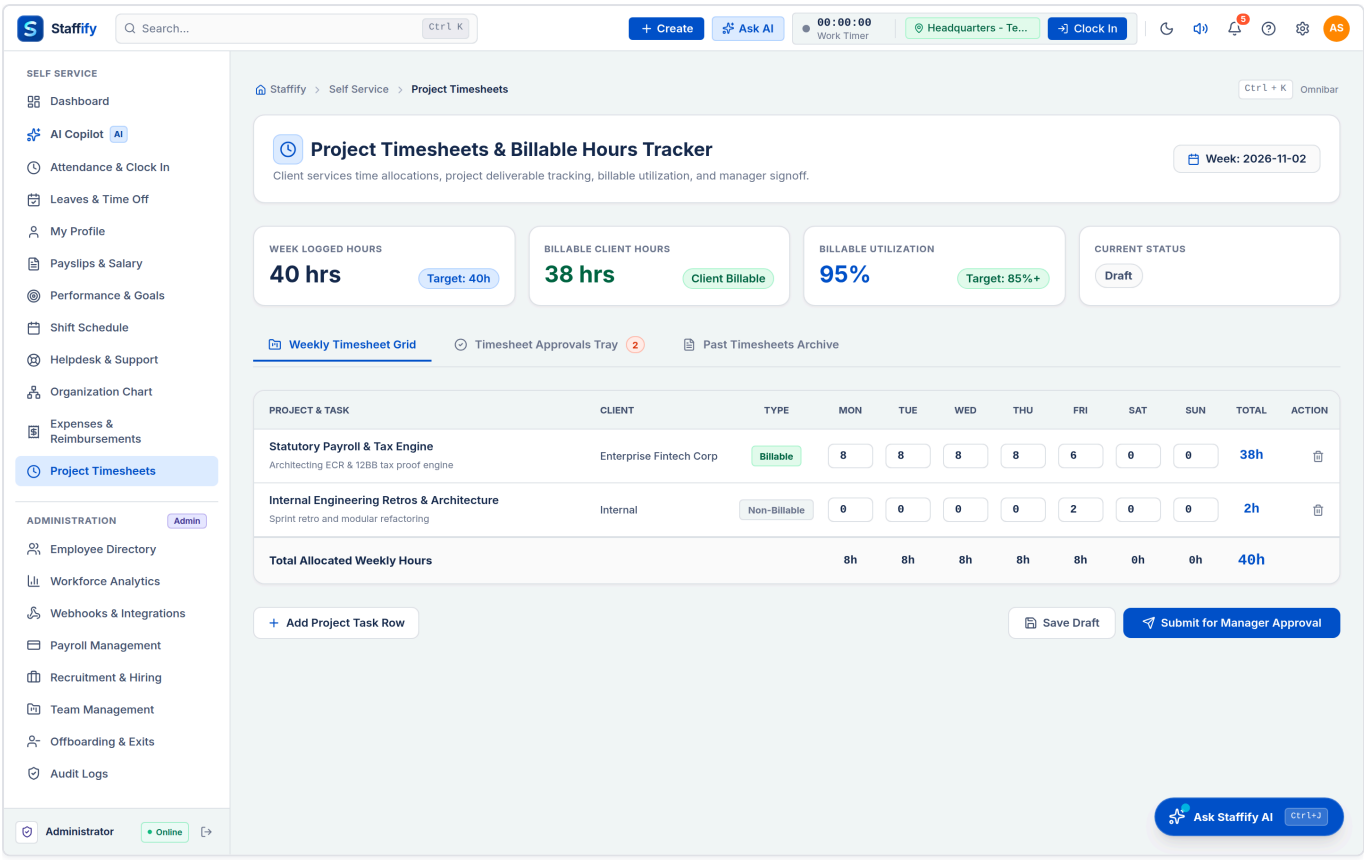


Figure: Weekly Client Timesheets

Workflows

- **Day-Wise Hours Entry**: Log Monday through Friday project hours against assigned client accounts.
- **Weekly Approval Cycle**: Employees submit completed timesheets at the end of each week for supervisor sign-off.
- **Overtime Calculation**: Identifies billable overtime hours exceeding 40 hours per week.

Module 13: Interactive Organization Hierarchy Tree

Dynamic visual organization chart rendering company-wide reporting relationships and squad structures.

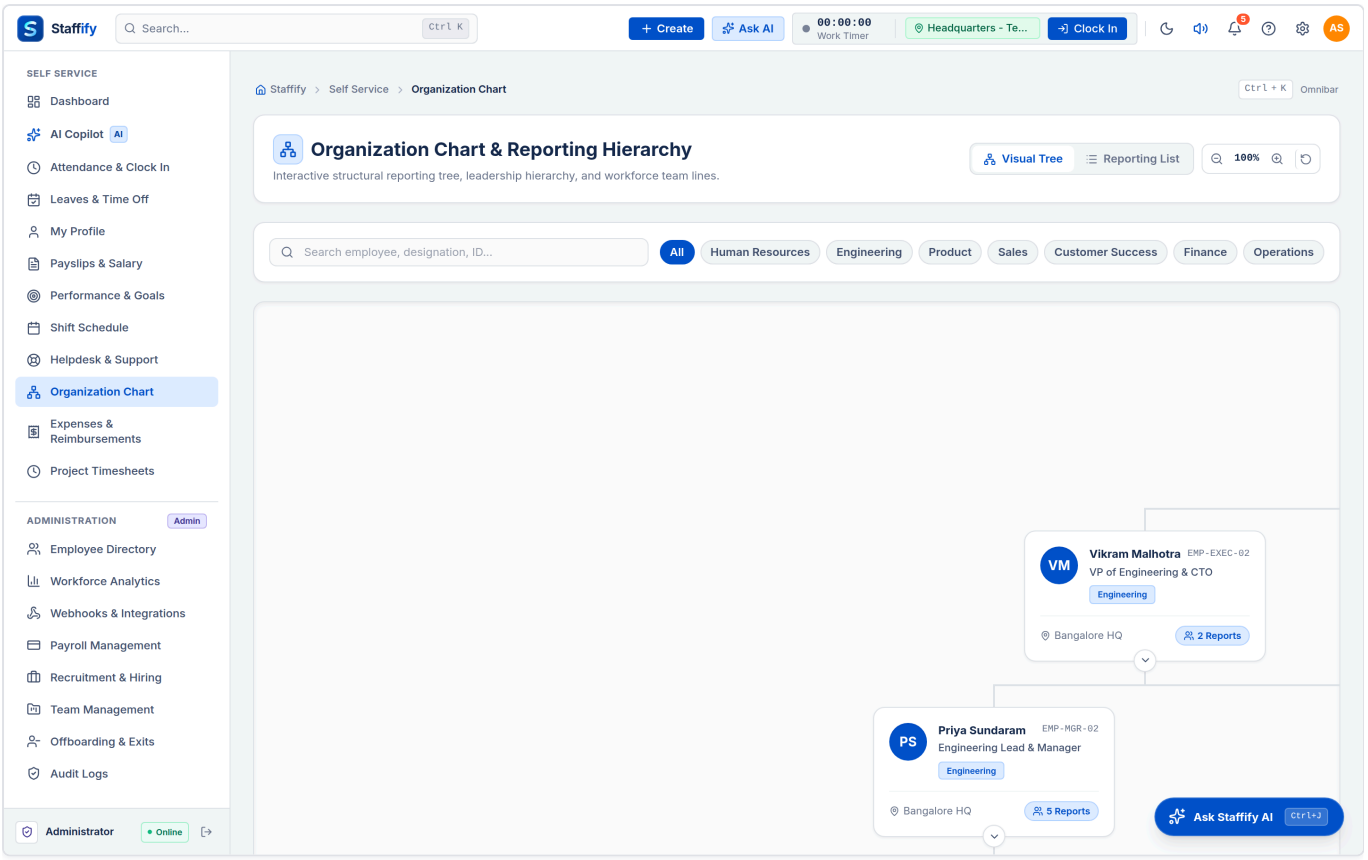


Figure: Interactive Organization Hierarchy Tree

Features

- **Expandable Node Hierarchy**: Click on any executive or team lead to expand direct and indirect reporting trees.
- **Node Profile Badges**: Displays designations, departments, employee IDs, and quick contact cards.
- **Zoom & Pan Canvas**: Fluid navigation across small teams or large 500+ employee structures.

Module 14: Executive Workforce Analytics & Retention Reporting

Strategic human capital analytics dashboard providing executive insights into workforce dynamics.

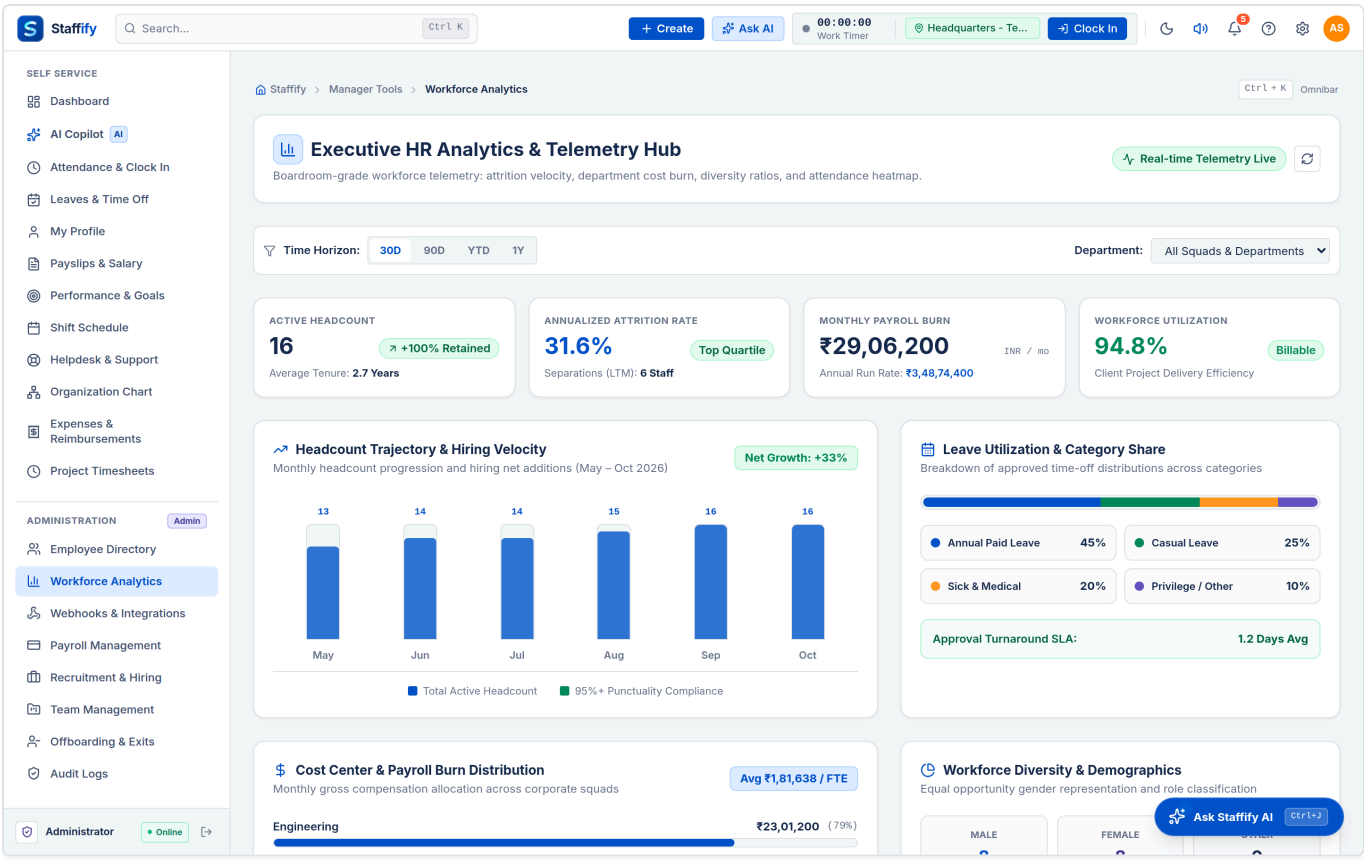


Figure: Executive Workforce Analytics

Metrics Covered

- **Headcount by Department**: Interactive breakdown across Engineering, HR, Product, Sales, and Finance.
- **Attrition & Retention Trends**: 12-month rolling retention rates and voluntary departure patterns.
- **Gender & Diversity Ratios**: Organization-wide diversity tracking.
- **Leave Utilization Rates**: Identifies underutilized and overutilized leave categories across squads.

Module 15: Employee Offboarding & Full-and-Final (F&F) Settlement

Structured employee departure lifecycle guaranteeing thorough departmental clearances and compliant payouts.

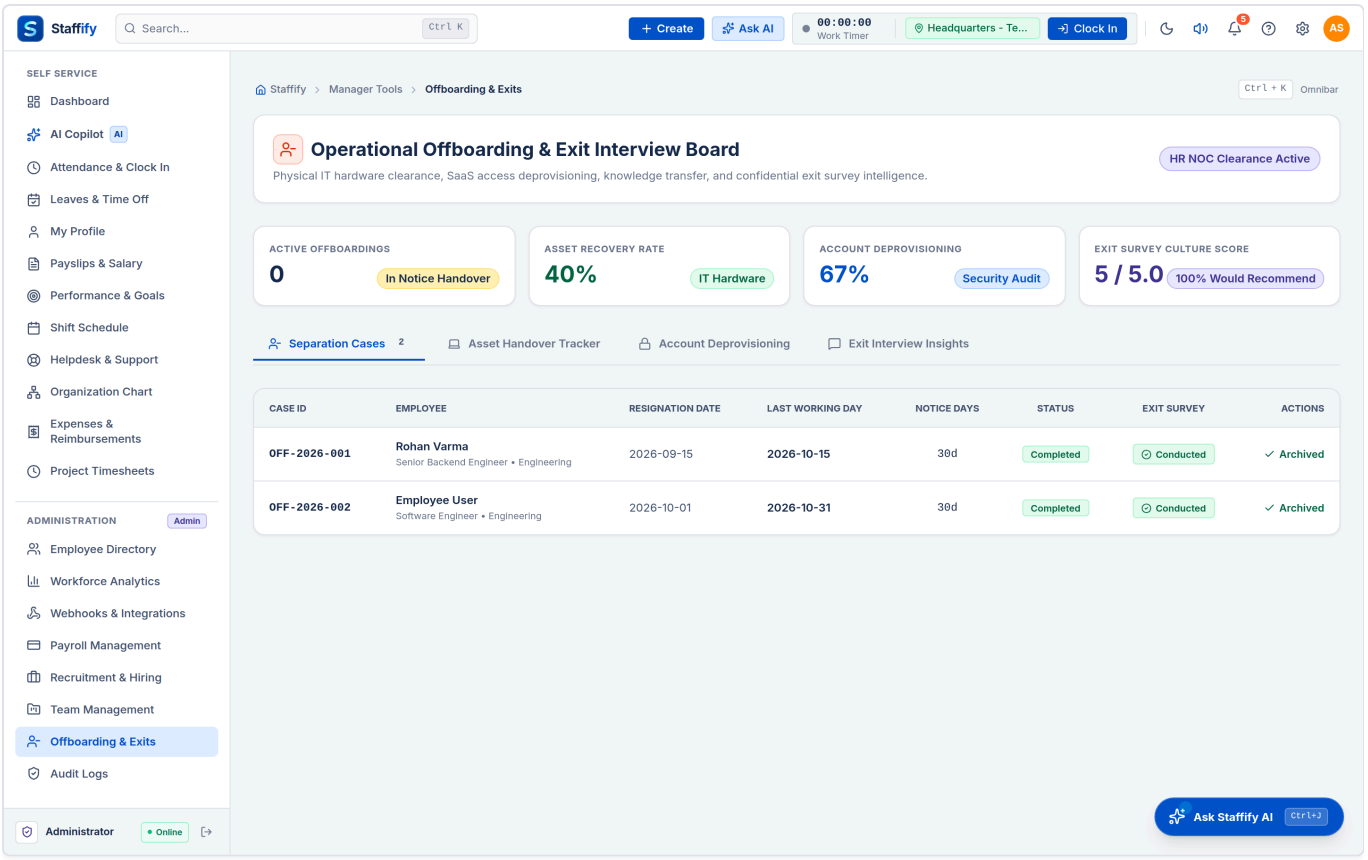


Figure: Employee Offboarding & F&F Settlement

Multi-Department Clearance Checklist

Every resignation triggers an automated clearance workflow across: 1. **IT Department**: Hardware asset recovery, account deactivation, and access revocation. 2. **Finance Department**: Outstanding salary advances/loan recovery and travel expense reconciliation. 3. **Operations & Facilities**: Desk handover, physical access keycard return. 4. **HR Department**: Exit interview completion and final gratuity/leave encashment calculation.

Full & Final Payout Execution

- Computes unencashed leave balances (EL encashment formula: `Basic / 30 Unused Days`).
- *Calculates statutory gratuity for employees completing 5+ years of continuous service.*
- *Generates official Relieving and Experience certificates.*

Module 16: Staffify AI Copilot & Multimodal Document Studio

Centralized enterprise AI workspace combining verified database intelligence with legal HR document generation.

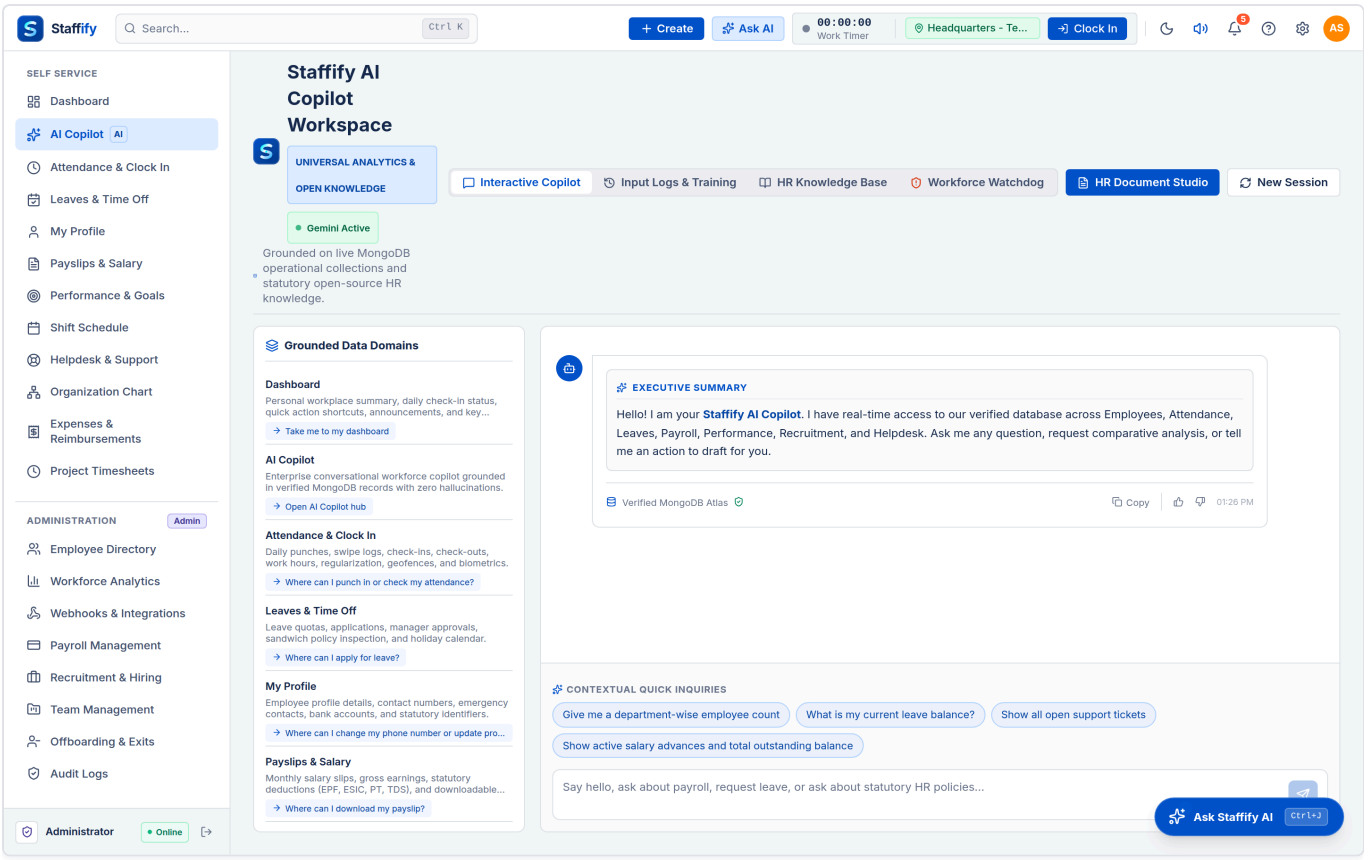


Figure: Staffify AI Copilot & Document Studio

Dual-Tab Capabilities

- ***Tab 1: AI Copilot Assistant***:

- Conversational natural language queries backed by our Deterministic Semantic Query Engine (DSQE). - Verifies database records in real time with zero hallucination. - Suggests interactive module navigation cards and pre-drafted authorized actions.

- ***Tab 2: Multimodal Document Studio***:

- Auto-synthesizes standardized, legally grounded Appointment Letters, Relieving Letters, Bonafide Certificates, and Salary Revision Letters. - Generates downloadable print-ready documents with official letterhead and authorized signatory seals.

Module 17: Tamper-Evident System Audit Trail Logs

Enterprise compliance audit log capturing all security-sensitive actions across the platform.

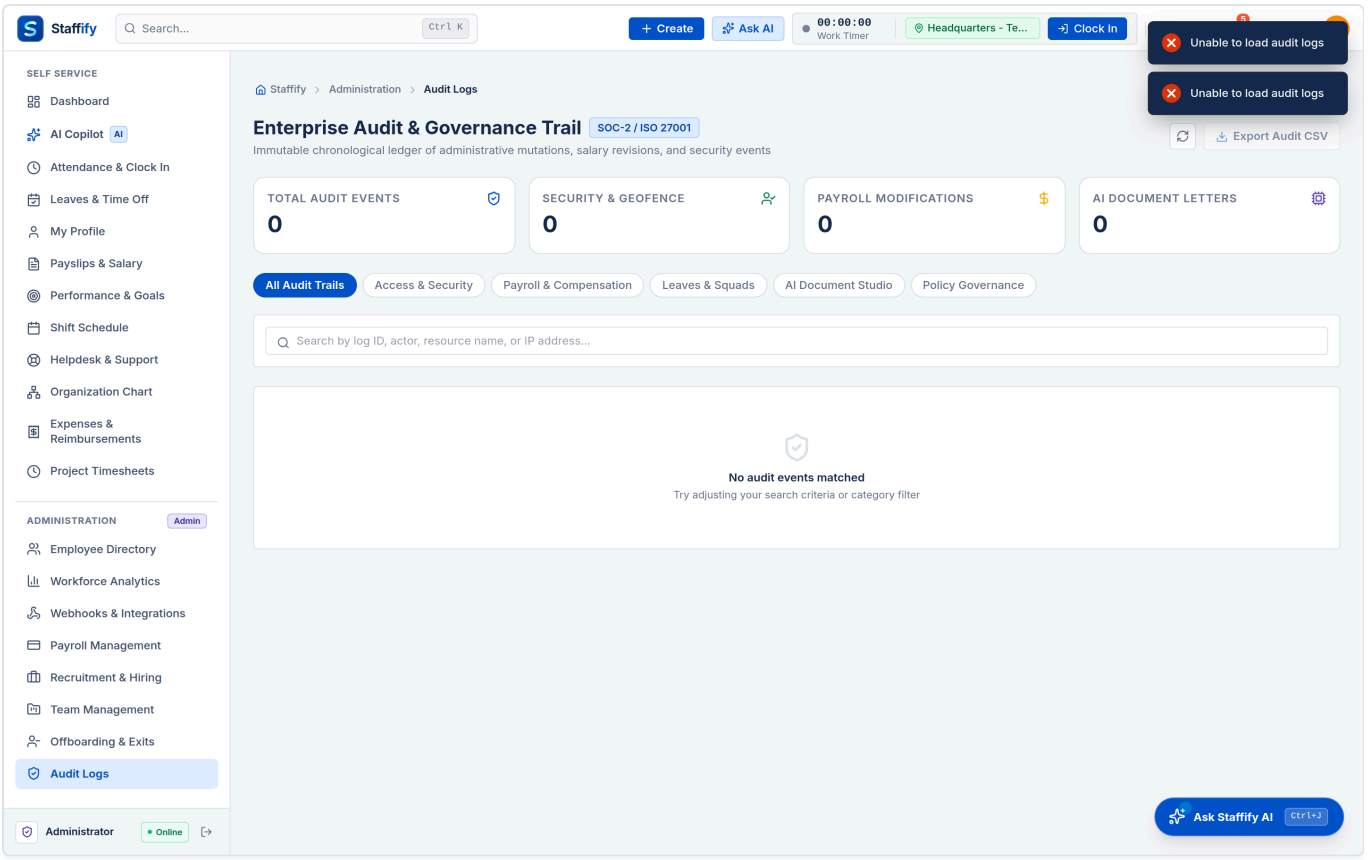


Figure: Tamper-Evident System Audit Trail Logs

Recorded Event Metadata

- ***Timestamp**: Precise millisecond ISO timestamp.*
- ***Actor Identity**: User ID, name, and role initiating the event.*
- ***Action Type**: E.g. `LOGIN_SUCCESS`, `PAYROLL_LOCKED`, `EMPLOYEE_MUTATED`, `EXPENSE_APPROVED`.*
- ***IP Address & User Agent**: Client origin IP and browser signature.*
- ***Target Record**: Affected database collection and entity identifier.*

Module 18: Webhook Bridge & Outbound Event Integrations

Outbound integration gateway broadcasting platform lifecycle events to third-party endpoints (Slack, Microsoft Teams, Zapier).

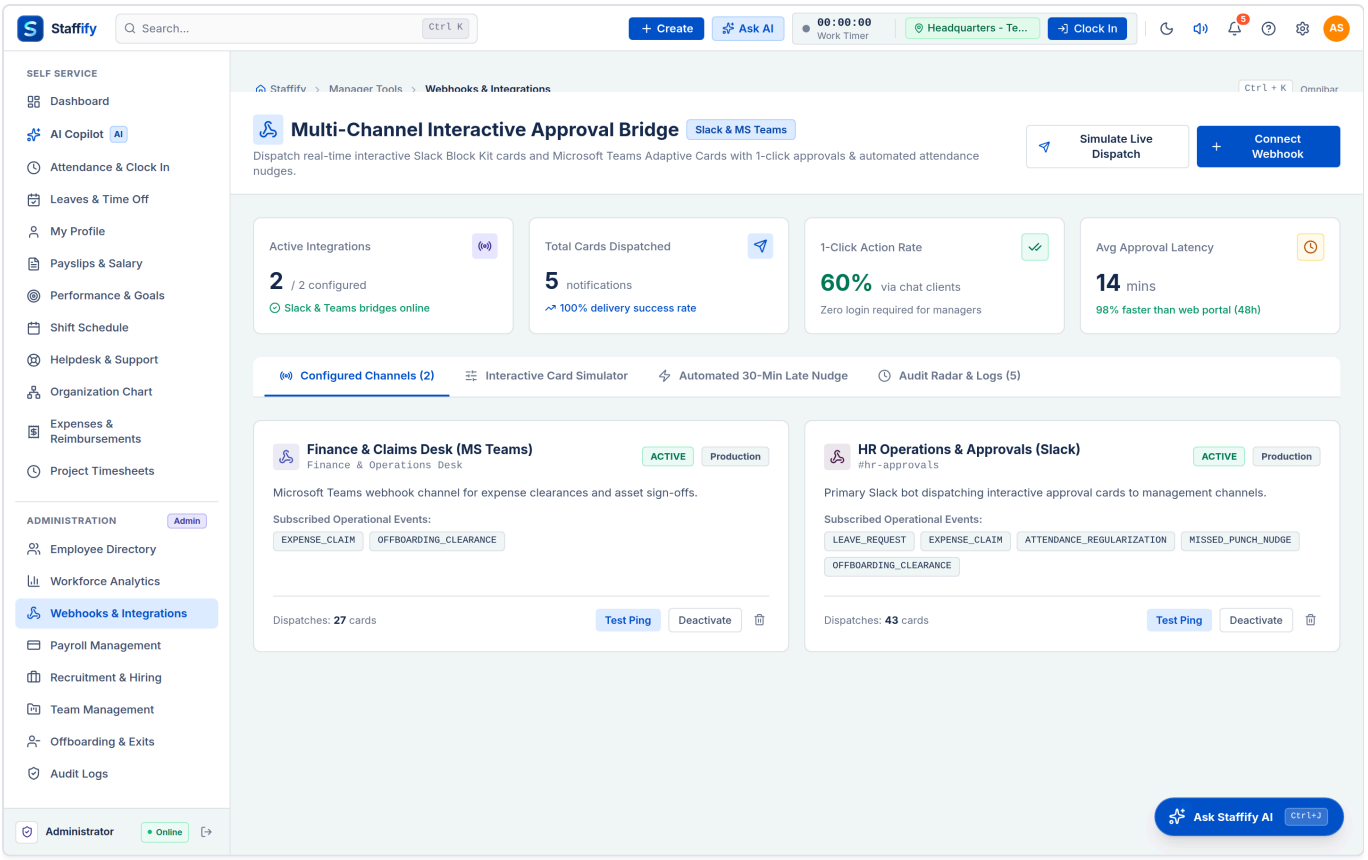


Figure: Webhook Bridge Integration Gateway

Event Triggers

- `attendance.clock_in` / `attendance.clock_out`
- `leave.applied` / `leave.approved`
- `payroll.disbursed`
- `candidate.hired`
- `expense.approved`
- `helpdesk.ticket_created`

Security Standards

- *HMAC-SHA256 signature generated using a shared secret header (`X-Staffify-Signature`).*
- *Exponential backoff retry queue for transient HTTP delivery failures.*

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22. Universal Zero-Popup Print & Vector PDF Engine

Staffify implements a zero-popup print and vector PDF architecture, replacing popup-blocked `window.open` *calls with a clean hidden*

